Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python services, concurrent systems, simulations, and reliable production workflows. Experienced in C++, Java, Linux, algorithms, operating systems, Docker, Kubernetes, unit and integration testing, debugging, performance analysis, and operational support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs and Azure SQL services for 500+ users, profiling indexed queries and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, deterministic recovery, maintainability, operational resilience, and systematic production anomaly diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, integrating secure document workflows, authentication, status tracking, and operational controls while systematically debugging production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python and PyTorch simulation pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured evaluation, performance monitoring, validation checks, and systematic debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and scalable REST integrations using AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and data-processing efficiency 68% across applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect investigation, release verification, and reliable deployments across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, securely processing 2 million+ Azure SQL records through documented software interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving cloud reliability, deployment consistency, and operational visibility.

Skills

Languages & Systems: C++, Python, Java, JavaScript, TypeScript, SQL, object-oriented programming, data structures, algorithms, operating systems, concurrency, multithreading
Simulation & Services: simulation, deterministic workloads, performance analysis, PyTorch, Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, distributed workflows
Platforms & Data: Linux, Windows, macOS, Docker, Kubernetes, AWS, Azure, S3, AWS RDS, Azure SQL, PostgreSQL, MySQL, MongoDB, cloud services
Quality & Operations: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, functional testing, automated testing, debugging, profiling, logging, monitoring, failure recovery, code review, documentation, production support, Agile

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Operating Systems*
- Built a C++ simulator implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance testing, reducing average turnaround time 25% across 1,000+ process configurations.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built a cross-platform workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, and Docker-backed deployment across modular Java and TypeScript services on multiple operating systems.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time WebSocket synchronization, MongoDB persistence, Kubernetes deployment, operational monitoring, and automated release workflows.